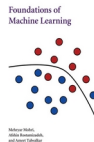
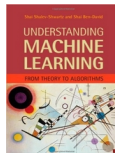
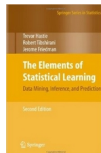
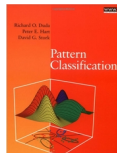
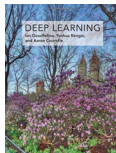
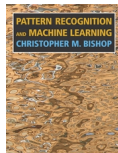


Intro to Machine Learning

Textbook and References

- Many excellent texts but none “required”. Some of them include (list not exhaustive)



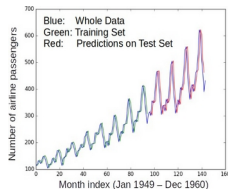
- Different books might vary in terms of
 - Set of topics covered
 - General approach taken e.g., classical statistics, deep learning, probabilistic/Bayesian, theory
- Terminology and notation (beware of this especially)**
- Avoid using too many sources until you have developed a reasonable understanding of a concept
- We will provide you the reading material from the relevant sources

Machine Learning (ML)

- Designing algorithms that ingest data and learn a (hypothesized) model of the data

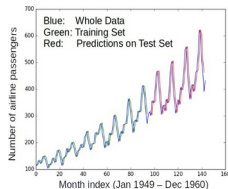
Machine Learning (ML)

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- data The learned model can be used to
 - Detect patterns/structures/themes/trends etc. in the
 - data Make predictions about future data and make decisions



Machine Learning (ML)

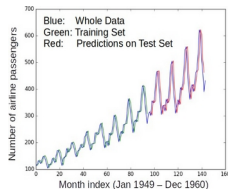
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- Modern ML algorithms are heavily “data-driven”

Machine Learning (ML)

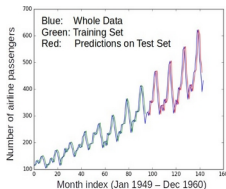
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 - No need to pre-define and hard-code all the rules (usually infeasible/impossible anyway)

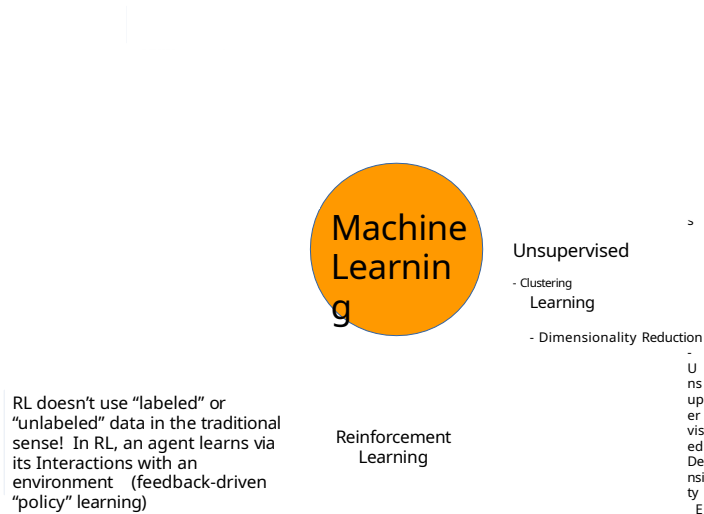
Machine Learning (ML)

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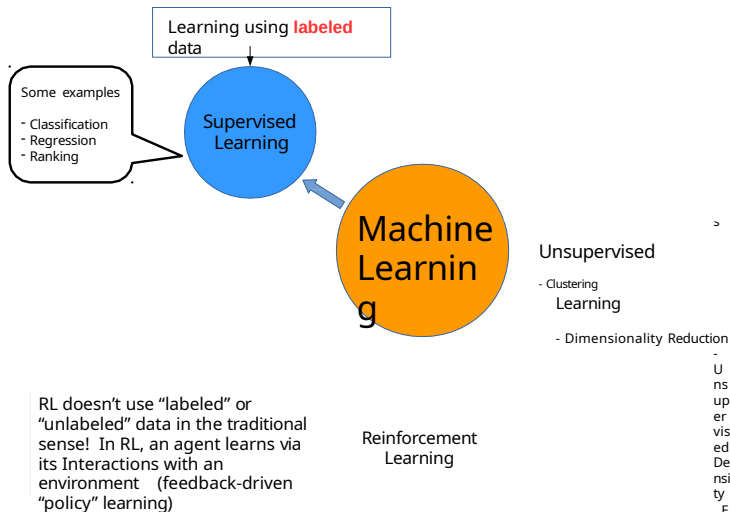


- Modern ML algorithms are heavily "data-driven"
 - No need to pre-define and hard-code all the rules (usually infeasible/impossible anyway) The rules are not "static"; can adapt as the ML algo ingests more and more data

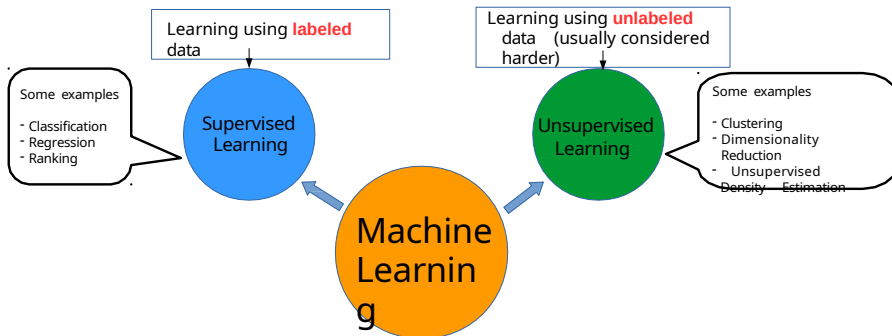
A Loose Taxonomy for ML



A Loose Taxonomy for ML



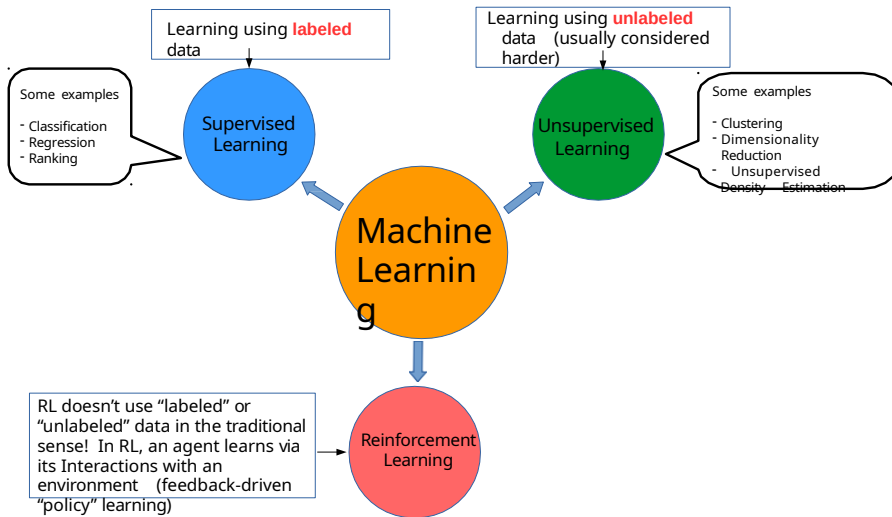
A Loose Taxonomy for ML



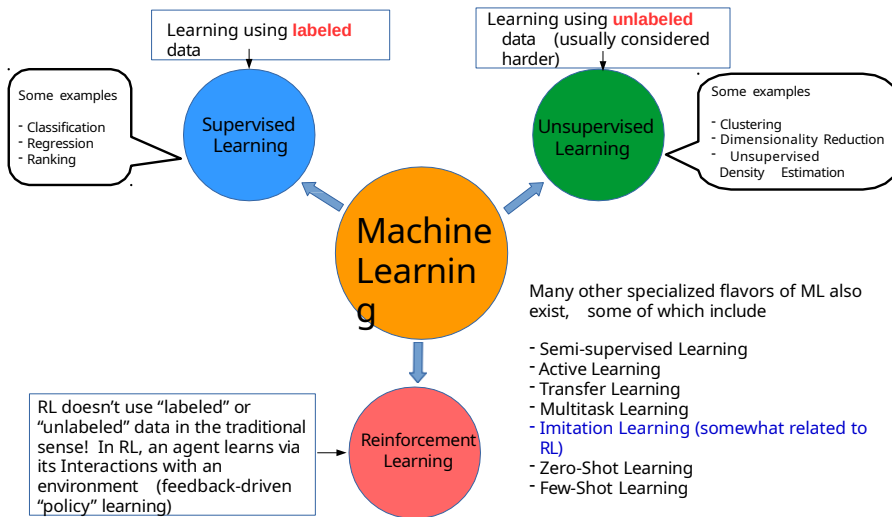
RL doesn't use "labeled" or "unlabeled" data in the traditional sense! In RL, an agent learns via its Interactions with an environment (feedback-driven "policy" learning)

Reinforcement Learning

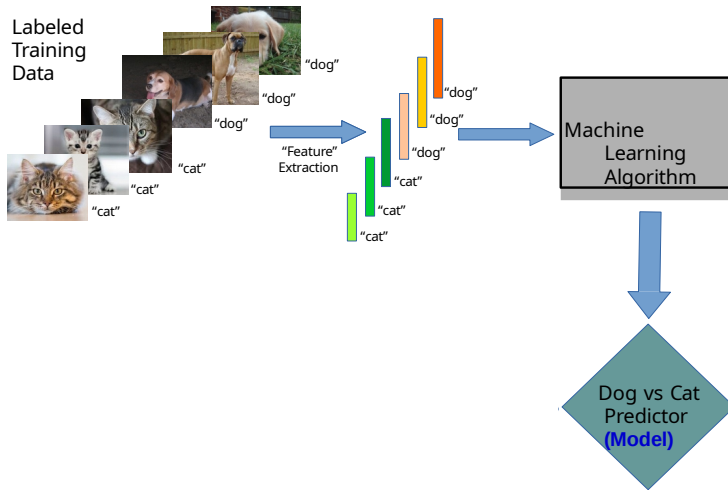
A Loose Taxonomy for ML



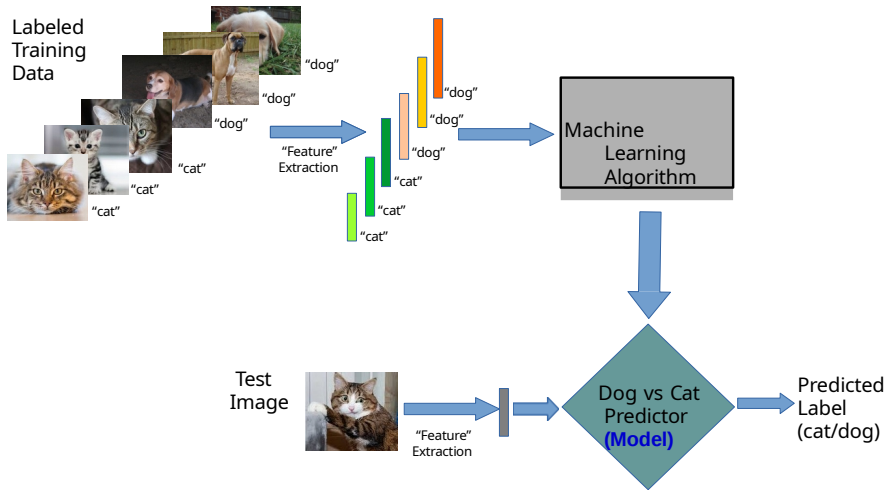
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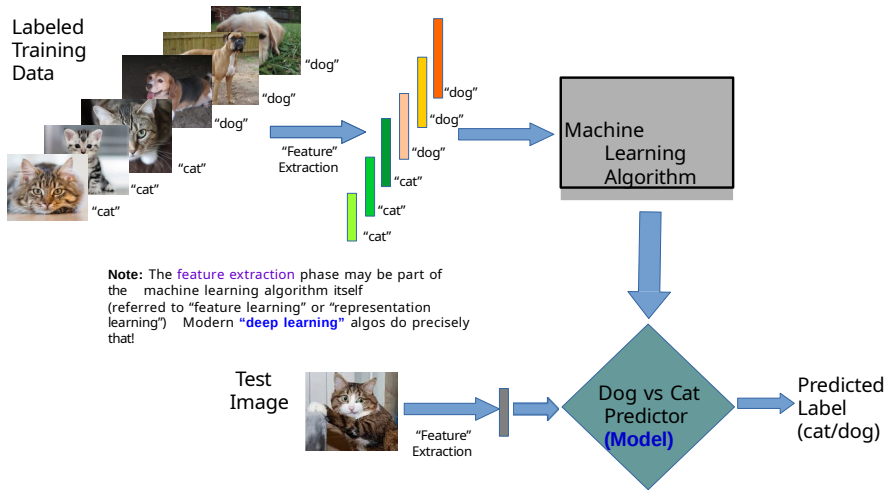
A Typical Supervised Learning Workflow (for Classification)



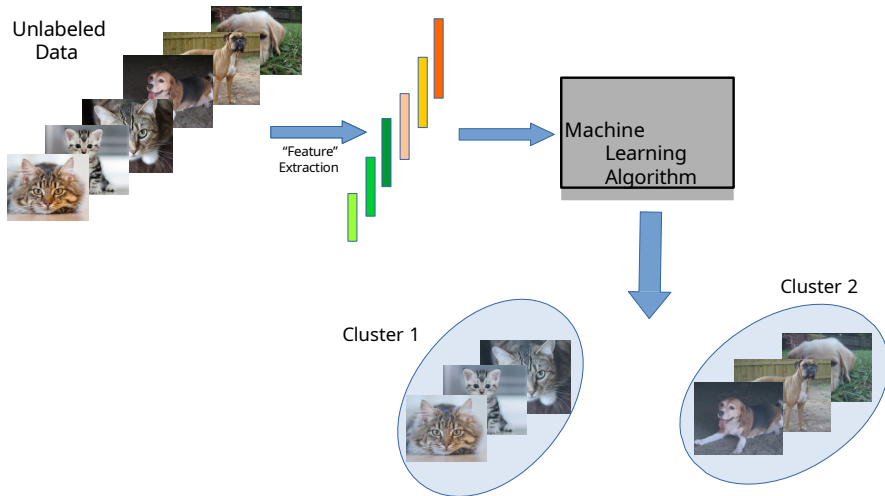
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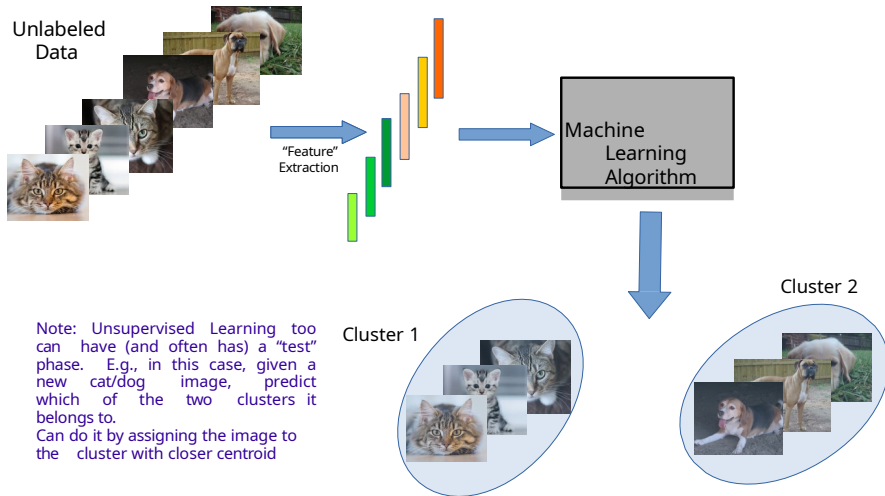
A Typical Supervised Learning Workflow (for Classification)



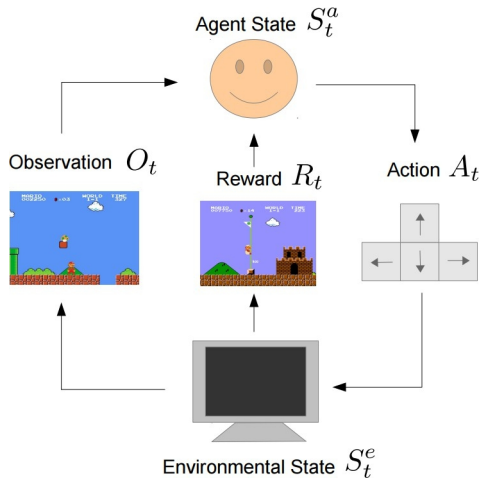
A Typical Unsupervised Learning Workflow (for Clustering)



A Typical Unsupervised Learning Workflow (for Clustering)



A Typical Reinforcement Learning Workflow

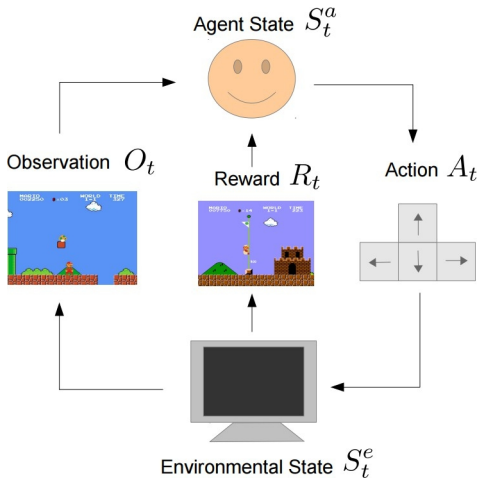


Agent's goal is to learn a policy for some task

Agent does the following repeatedly

- Senses/observes the environment
- Takes an action based on its current policy
- Receives a reward for that action
- Updates its policy

A Typical Reinforcement Learning Workflow



Agent's goal is to learn a policy for some task

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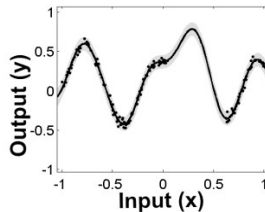
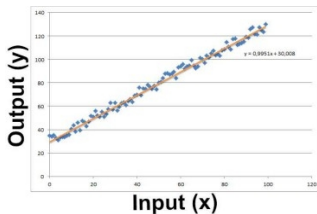
There IS supervision, not explicit (as in Supervised Learning) but rather implicit (feedback based)

Geometric View of Some Basic ML Problems

Regression

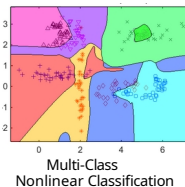
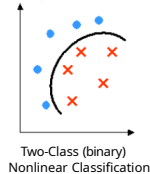
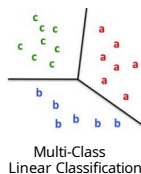
Supervised Learning: Learn a line/curve (the "model") using training data consisting of Input-output pairs (each output is a real-valued number)

Use it to predict the outputs for new "test" inputs



Classification

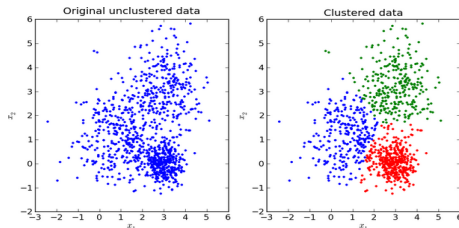
Supervised Learning: Learn a linear/nonlinear separator (the "model") using training data consisting of input-output pairs (each output is discrete-valued "label" of the corresponding input)
Use it to predict the labels for new "test" inputs



Geometric View of Some Basic ML Problems

Clustering

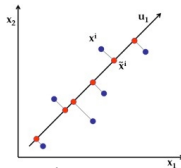
Unsupervised Learning: Learn the grouping structure for a given set of unlabeled inputs



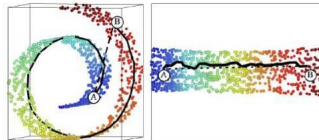
Dimensionality Reduction

Unsupervised Learning: Learn a Low-dimensional representation for a given set of high-dimensional inputs

Note: DR also comes in supervised flavors (supervised DR)



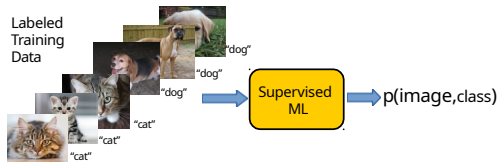
Two-dim to one-dim linear projection



Three-dim to two-dim nonlinear projection (a.k.a. manifold learning)

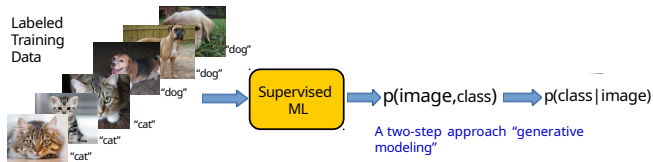
Machine Learning = Probability Density Estimation

- Supervised Learning (“predict y given x ”) can be thought of as estimating $p(y|x)$



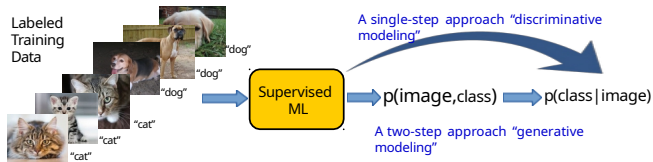
Machine Learning = Probability Density Estimation

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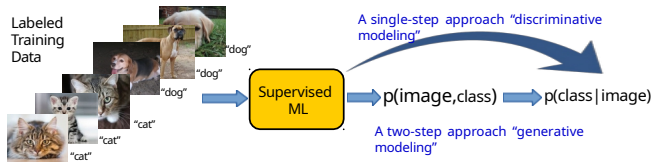
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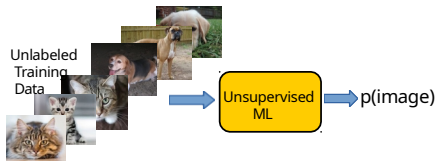


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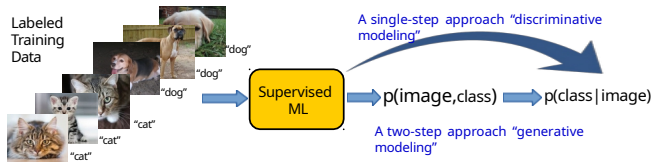


- Unsupervised Learning (“model x ”) can also be thought of as estimating $p(x)$

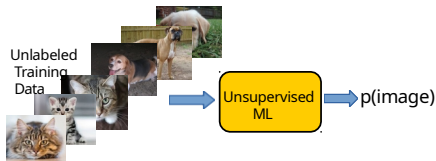


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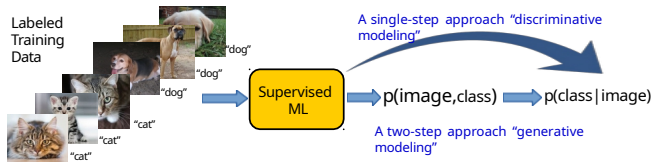
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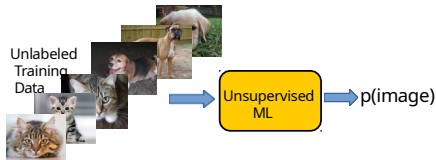
- Harder for Unsupervised Learning because there is no supervision y

Machine Learning = Probability Density Estimation

- Supervised Learning ("predict y given x ") can be thought of as estimating $p(y|x)$



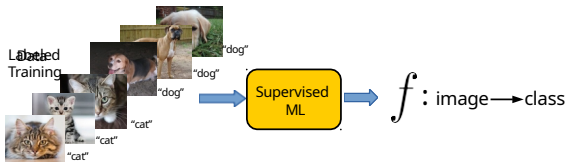
- Unsupervised Learning ("model x ") can also be thought of as estimating $p(x)$



- Harder for Unsupervised Learning because there is no supervision y
- Other ML paradigms (e.g., Reinforcement Learning) can be thought of as learning prob. density

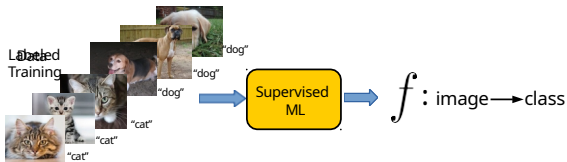
Machine Learning = Function Approximation

- Supervised Learning (“predict y given x ”) can be thought learning a function that maps x to y

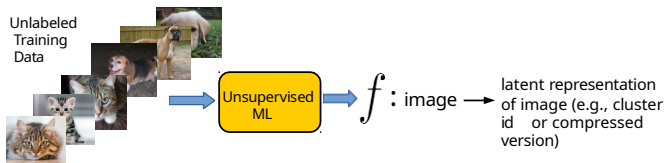


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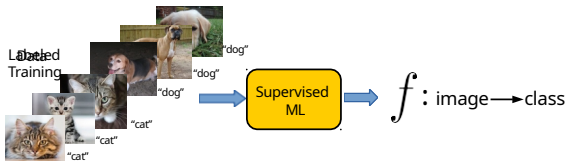


- Unsupervised Learning (“model x ”) can also be thought of as learning a function that maps x to some useful **latent representation** of x

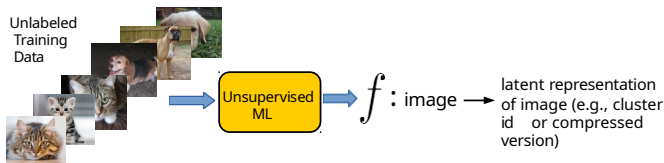


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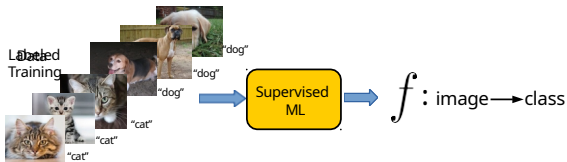
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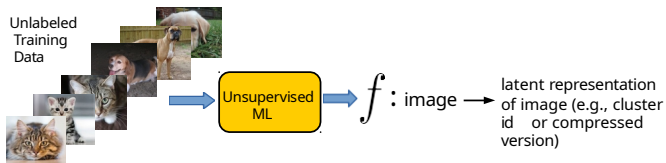
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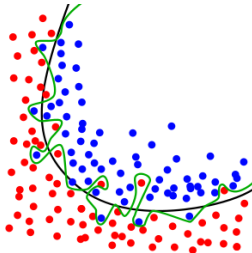
- Unsupervised Learning ("model x ") can also be thought of as learning a function that maps x to some useful **latent representation** of x



- Harder for Unsupervised Learning because there is no supervision y
- Other ML paradigms (e.g., Reinforcement Learning) can be thought of as doing function approx

Overfitting and Generalization

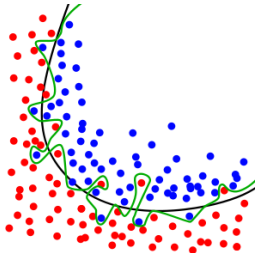
- Doing well on the training data is not enough for an ML algorithm



- Trying to do too well (or perfectly) on training data may lead to bad “generalization”

Overfitting and Generalization

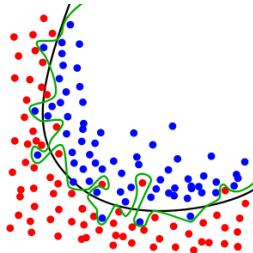
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- Generalization: Ability of an ML algorithm to do well on future “test” data

Overfitting and Generalization

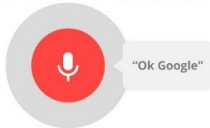
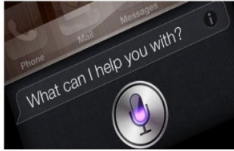
- Doing well on the training data is not enough for an ML algorithm



- Trying to do too well (or perfectly) on training data may lead to bad “generalization”
- Generalization: Ability of an ML algorithm to do well on future “test” data
- Simple models/functions tend to prevent overfitting and generalize well: A key principle in designing ML algorithms (called “regularization”; more on this later)

Machine Learning in the real-world

Broadly applicable in many domains (e.g., internet, robotics, healthcare and biology, computer vision, NLP, databases, computer systems, finance, etc.).



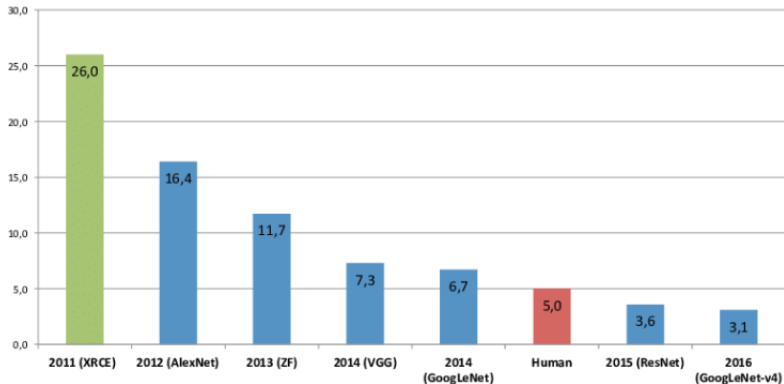
Predictive Policing



Online Fraud Detection

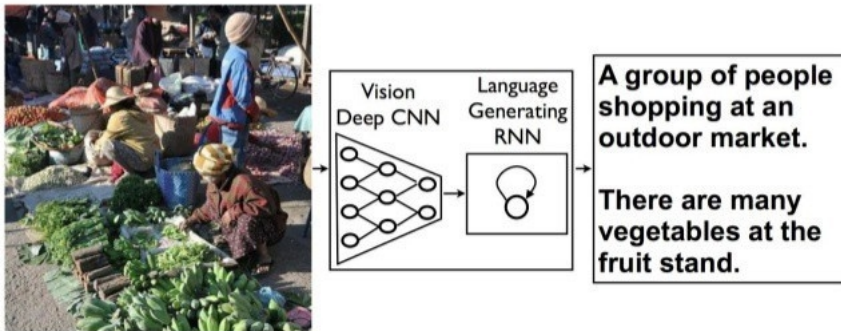
Machine Learning helps Computer Vision

ML algorithms can learn to recognize images better than humans!



Machine Learning helps Computer Vision

ML algorithms can learn to generate captions for images



<http://arxiv.org/abs/1411.4555> "Show and Tell: A Neural Image Caption Generator"

Machine Learning helps Computer Vision

ML algorithms can learn to answer questions about images (Visual QA)



What vegetable is on the plate?

Neural Net: **broccoli**

Ground Truth: broccoli



What color are the shoes on the person's feet ?

Neural Net: **brown**

Ground Truth: brown



How many school busses are there?

Neural Net: **2**

Ground Truth: 2



What sport is this?

Neural Net: **baseball**

Ground Truth: baseball



What is on top of the refrigerator?

Neural Net: **magnets**

Ground Truth: cereal



What uniform is she wearing?

Neural Net: **shorts**

Ground Truth: girl scout



What is the table number?

Neural Net: **4**

Ground Truth: 40



What are people sitting under in the back?

Neural Net: **bench**

Ground Truth: tent

Machine Learning helps NLP

ML algorithms can learn to translate text



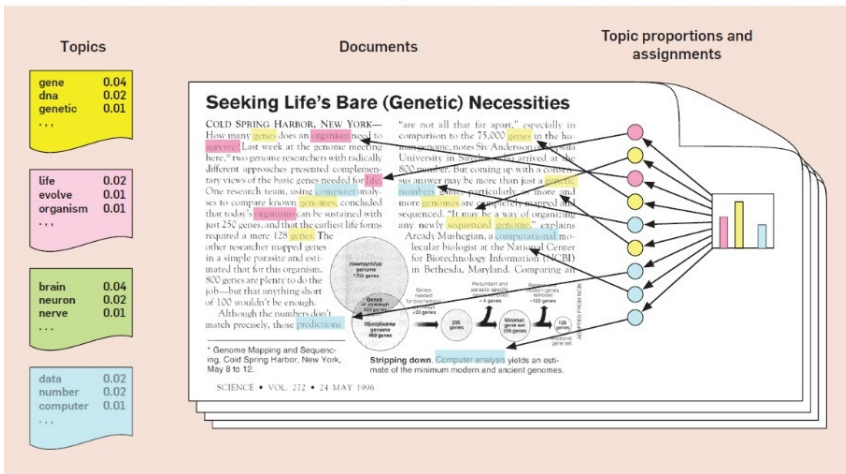
Machine Learning helps NLP

ML algorithms can learn to summarize text

Input: Article 1st sentence	Model-written headline
metro-goldwyn-mayer reported a third-quarter net loss of dlr 16 million due mainly to the effect of accounting rules adopted this year	mgm reports 16 million net loss on higher revenue
starting from july 1, the island province of hainan in southern china will implement strict market access control on all incoming livestock and animal products to prevent the possible spread of epidemic diseases	hainan to curb spread of diseases
australian wine exports hit a record 52.1 million liters worth 260 million dollars (143 million us) in september, the government statistics office reported on monday	australian wine exports hit record high in september

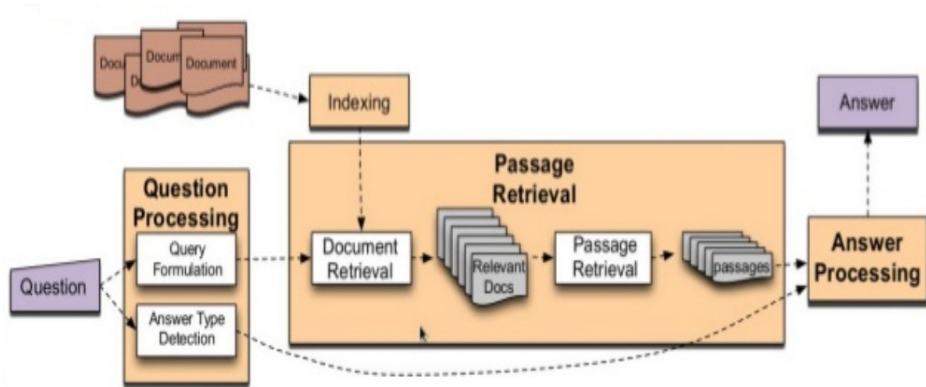
Machine Learning helps NLP

ML algorithms can learn the topics in a text corpus (“Topic Modeling”)



Machine Learning helps Search and Info Retrieval

ML algorithms can learn to search for the answer to a given question from a large database of documents



Machine Learning meets Speech Processing

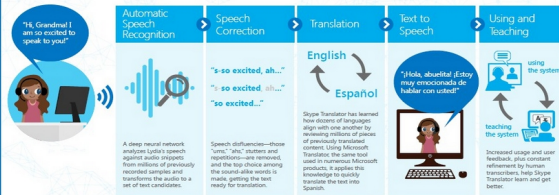
ML algorithms can learn to translate speech in **real time**

NOW YOU'RE SPEAKING MY LANGUAGE (LITERALLY)



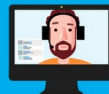
Skype has always been about making it easy to talk with family and friends all over the world. Now, by integrating advanced speech recognition and automatic translation into Skype, Skype Translator lets you speak with those you've always wished you could, even if they speak a different language.

HOW SKYPE TRANSLATOR WORKS



TRANSLATE INSTANT MESSAGES OVER 40 LANGUAGES

Holding a translated IM conversation is super easy: Choose a contact, turn on the translation switch for that person, and start typing. When you hit enter (or tap send), your original message will appear in the right-hand pane, followed by its translation. Your contact on the other end will see something very similar, albeit with the translated message in his/her preferred language presented first. While voice translation initially supports English and Spanish only, IM translation supports over 40 languages, so feel free to experiment with them all—even Klingon!



Register for the preview at www.skype.com/translator and wait for your invite.

Install the Skype Translator client.

Use Skype Translator to call someone who speaks Spanish. Or, if you speak Spanish, call someone who speaks English.

Every call you make helps Skype Translator get a little bit better. You won't see the improvement right away, but you will see gradual improvement over time.

PUTTING MACHINE LEARNING TO THE TEST

To provide a seamless user experience, Skype Translator uses machine learning to solve key challenges in interpreting human language, including:



Representing the different ways people really speak



Determining sentence boundaries, punctuation and case from speech



Disambiguating sound-alike words in context



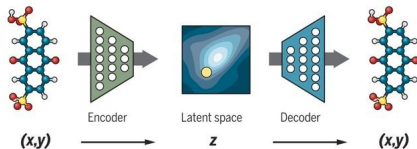
Mapping words and phrases from one language to another



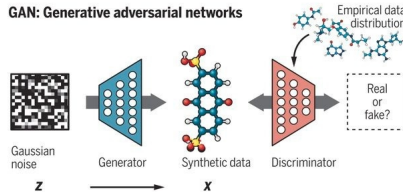
Machine Learning helps Chemistry

ML algorithms can understand properties of molecules and learn to synthesize new molecules

VAE: Variational autoencoders

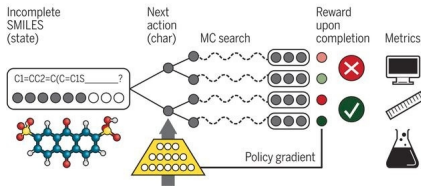


GAN: Generative adversarial networks

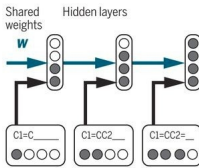


RL: Reinforcement learning

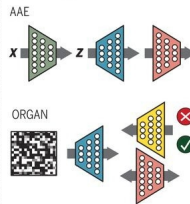
Policy gradient with Monte Carlo tree search (MCTS)



RNN: Recurrent neural network

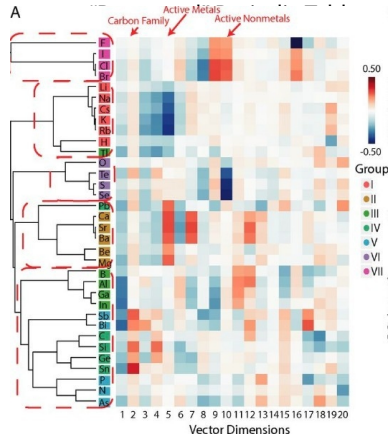
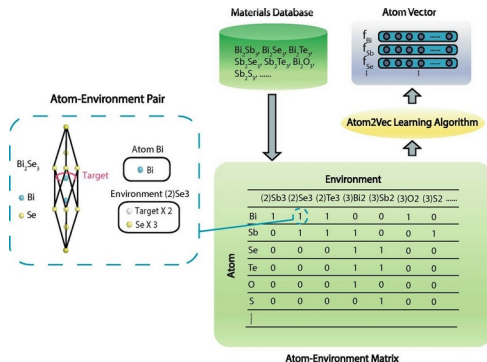


Hybrid approaches



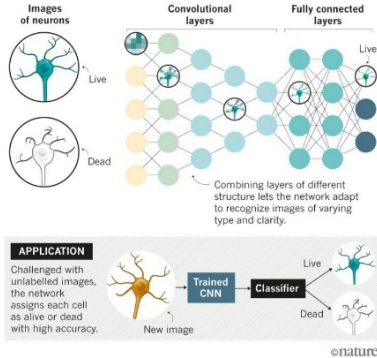
Machine Learning helps Chemistry

ML algorithms can “read” databases of materials and **recreate the Periodic Table within hours**



Machine Learning helps Many Other Areas..

Biology



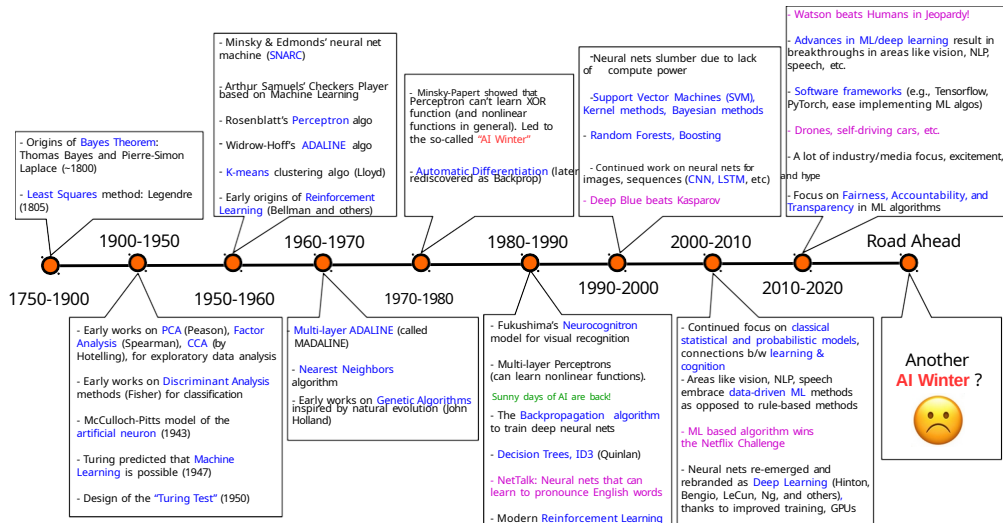
Source: Jeremy Linley/Drew Linley/Steve Finkbeiner/Thomas Seme

Finance



Picture courtesy: (1) <https://www.nature.com/articles/d41586-018-02174-z> (2)

Machine Learning: A Brief Timeline and Some Milestones



(Tentative) List of topics

- Supervised Learning
 - nearest-neighbors methods, decision trees
 - linear/non-linear regression and classification
- Unsupervised Learning
 - Clustering and density estimation
 - Dimensionality reduction and manifold learning Latent factor models and matrix factorization
- Probabilistic
- Modeling Deep
- Learning Ensemble
- Methods

Course Goals

By the end of the semester, you should be able to:

- Understand how various machine learning algorithms work
- Implement them (and, hopefully, their variants/improvements) on your own
- Look at a real-world problem and identify if ML is an appropriate solution
- If so, identify what types of algorithms might be applicable
- Feel inspired to work on and learn more about Machine Learning :-)

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By the end of the semester, you should be able to:

- **Understand** how various machine learning algorithms work
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- Look at a real-world problem and **identify** if ML is an appropriate solution
- If so, identify what types of algorithms might be applicable
- **Feel inspired** to work on and **learn more** about Machine Learning :-)

Caution: There will be quite a bit of maths in this course (can't be avoided!). You are expected to be (or to make yourself) comfortable with multivariate calculus, linear algebra, probability and statistics. Please use the provided reference materials to brush up these concepts.